

Input Tokens

t	0	1	2	3	4	5
	3	3	0	3	0	2
	2	0	3	2	3	1
	1	0	1	1	1	3
	3	1	2	3	1	1
	2	3	2	2	1	2
	↓	↓	↓	↓	↓	↓

Multi-head Self-Attention

Head 0

Linear

W_q

1 -1 1 0 0 →

W_k

1 0 0 0 0 →

W_v

0 1 0 0 0 →

Q

2 3 -2 2 -2 4

K

3 3 0 3 0 2

V

2 0 3 2 3 1

Scaled Dot-Product

t

0 1 2 3 4 5

Q

2 3 -2 2 -2 4

K^T

3 7 3

K^T * Q / sqrt(d_k)

6.35 15.6 -7.5 6.35 -9.2 5.2

V

2 0 3 2 3 1

Softmax

0.48 0.5 0.01 0.48 0 0.46

Head 1

Linear

W_q

1 -1 0 1 1 →

W_k

1 0 1 -1 1 →

W_v

1 0 -1 0 -1 →

Q

6 7 1 6 -1 4

K

3 5 1 3 1 6

V

0 0 -3 0 -2 -3

Scaled Dot-Product

t

0 1 2 3 4 5

Q

6 7 1 6 -1 4

K^T

3 6 10

K^T * Q / sqrt(d_k)

46.2 22.5 26 46.2 22.5 13.9

V

0 0 -3 0 -2 -3

Softmax

0.49 0 0.5 0.49 0.5 0

Head 2

Linear

W_q

0 -1 0 0 0 →

W_k

0 1 -1 0 0 →

W_v

-1 1 1 0 1 →

Q

-2 0 -3 -2 -3 -1

K

1 0 2 1 2 -2

V

2 0 6 2 5 4

Scaled Dot-Product

t

0 1 2 3 4 5

Q

-2 0 -3 -2 -3 -1

K^T

1 1 3

K^T * Q / sqrt(d_k)

-1.2 0.58 3.46 -1.2 1.15 -2.9

V

2 0 6 2 5 4

Softmax

0.03 0.31 0.26 0.03 0.13 0.02

Concat

Head	V'
0	1.94 2 2.65 1.94 2.63 1.83
	2.99 3 0.37 2.99 0.38 2.99
	0.9 1 -0.2 0.9 -0.2 0.74
1	-0 -2.9 -0 -0 -0 -2.7
	1.98 0.97 2 1.98 2 0.92
	2.98 3.88 3 2.98 3 3.63
2	0.12 1.56 1.06 0.12 0.52 0.07
	4.18 5.94 5.59 4.18 4.78 4.1
	0.06 0.75 0.53 0.06 0.26 0.03

Linear

	W _o	
0	0 0 1 0 0 1 -1 -1 -1	→
1	1 1 1 0 1 1 1 1 1	→
1	0 -1 1 0 0 1 1 0 0	→
1	0 0 0 0 0 -1 1 0 0	→
-1	1 1 1 1 0 0 -1 1 0	→

Output

↓	↓	↓	↓	↓	↓
-0.5	-3.4	-4.3	-0.5	-2.8	0.17
15.1	19.1	15	15.1	13.4	14.3
1.14	-0.3	3.88	1.14	3.35	-1.5
-0.9	-0.3	0.72	-0.9	0.15	-1.7
5.99	3.47	2.07	5.99	1.81	3.24

Output Tokens

